

GetSnap User Manual

Connect an instrument and capture screen images / waveform data with a single hotkey or button press, then save or copy them.

[Instrument] --Cable(LAN/USB/GPIB)--> [GetSnap] --Hotkey/Button--> [Image/Data] --> [Save to Folder / Clipboard]

The screen-capture commands are initial defaults based on each vendor's commonly published command set. They may differ by model/firmware. You can adjust them directly from the app — see "6. Adding & Editing Instruments" below.

1. Launching the App

Method	Action
exe file (recommended)	Double-click <code>dist\GetSnap.exe</code>
From Python	<code>venv\Scripts\activate</code> → <code>python app.py</code>

GPIB connections require NI-VISA or Keysight IO Libraries Suite (see "7. Troubleshooting").

2. Screen Layout

Tab	Contents
Capture	Capture buttons, Legend, DIV Conditions, preview, history
Devices	VISA resource list, connect
Settings	Save location, background mode, hotkeys, instruments, Legend, DIV Conditions, language

Save to Folder

Copy to Clipboard

Save Data (CSV, etc.)

Enable Legend (applies to folder save & clipboard copy)

MODEL	
Ope. Mode	
BOM	
Condition	
Note	

Enable measurement conditions (shows enabled channels' voltage range / timebase on the right of the image)

Enable auto-coloring (reads the actual display color from the instrument)

Legend to Display in the Image

CH1

Vds

CH2

Vgs

++ Add Channel

No Snap

History

Capture tab

Detected VISA Resources

USB0::0x0B21::0x0025::91W123456::INSTR

GPIB0::7::INSTR

 Refresh

 Connect

Manually enter a resource string (if not auto-detected)

e.g. GPIB0::7::INSTR

 Manual Connect

Connected: YOKOGAWA,DLM5058,91W123456,FW2.34
[Yokogawa]

Devices tab

3. Basic Operation

Step	Action
① Connect	"Devices" tab → "Refresh" → select target → "Connect"
② Capture	Hotkey, or the buttons on the "Capture" tab
③ Check	Review the preview/history on the "Capture" tab
④ Paste	Ctrl+V in Excel, etc. (when copied to clipboard)

Hotkeys

Purpose	Key
Save to folder	Ctrl+Alt+F1
Copy to clipboard	Ctrl+W

When an Error Occurs

The [①...] shown at the start of the error dialog indicates which step failed.

Step shown	Meaning
Instrument not connected / Preset not registered	Not yet connected on the "Devices" tab, or no preset is registered
①Connecting to VISA resource / ①Sending command / receiving response	Communication issue (cable, address, VISA runtime)
②Identifying model / ②Decoding image	Response data format issue
③Post-processing	Image processing issue (background conversion, Legend/DIV Conditions compositing)
④Saving / ④Copying	Permissions or disk space issue at the save destination

4. Adding Information to the Image: Legend & DIV Conditions

Two toggles on the Capture tab let you auto-composite information onto the image. Both apply to folder save and clipboard copy alike.

 Save Data (CSV, etc.)

☒ Enable Legend (applies to folder save & clipboard copy)

MODEL	
Ope. Mode	
BOM	
Condition	
Note	

☒ Enable measurement conditions (shows enabled channels' voltage range / timebase on the right of the image)

☒ Enable auto-coloring (reads the actual display color from the instrument)

Legend to Display in the Image

CH1	Vds		
CH2	Vgs		

 + Add Channel

Legend / DIV Conditions (Capture tab)

4.1 Legend: free-text fields attached to the image

Turning on "Enable Legend" shows a list of items on the Capture tab (e.g. MODEL, Ope. Mode, BOM, Condition, Note). Fill in values and capture — they're composited as a table at the top of the image. Adding/removing items, reordering, and font settings are managed from the "Legend" section of the

"Settings" tab (not covered by "6. Adding & Editing Instruments").

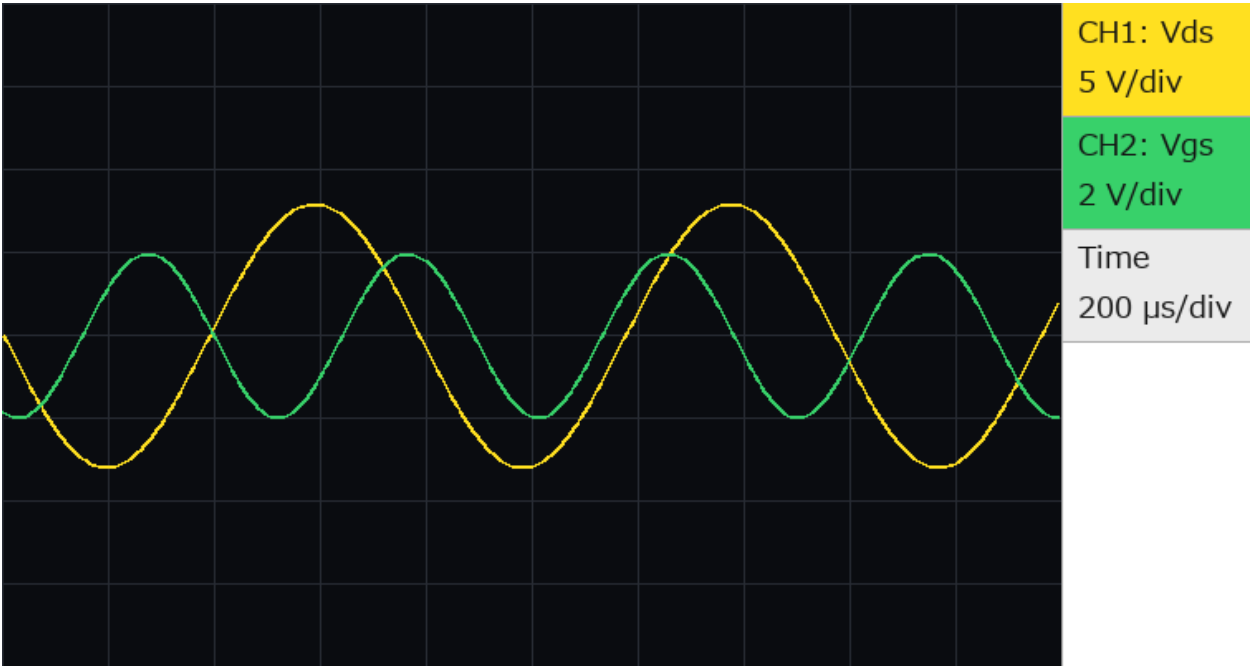
4.2 DIV Conditions: per-channel voltage range / timebase, shown automatically

Turning on "Enable measurement conditions" lets you pick which channels (CH1–8) / Math traces (M1–8) to display, and enter a custom legend name per row (e.g. CH1: Vds). On every capture, GetSnap queries the instrument and composites the actual voltage range / timebase (e.g. 5 V/div) into a panel whose row backgrounds match each channel's waveform color.

Action	Steps
Add a channel	" + Add Channel" → choose CH1–8 / M1–8 from the dropdown
Edit the legend name	Type into each row's text field (leave blank to show just the ID, e.g. "CH1")
Remove a channel	Trash-can icon at the right of each row
Change the insertion position	"Settings" tab → "DIV Conditions Position" (Right / Left / Bottom / Top)
Change the font	"Settings" tab → "DIV Conditions Font"

Choosing "Top" inserts the DIV Conditions panel between the Legend and the waveform image (stacking order: Legend → DIV Conditions → waveform).

Example output (CH1=Vds, CH2=Vgs shown on the right):



DIV Conditions panel example

4.3 Choosing channel colors (automatic or manual)

With "Enable auto-coloring" on, each panel row's background color is read from the instrument via a command (requires the color-query commands to be configured on the instrument preset — see "6.6 DIV Conditions commands"). With it off, click the color button (the square swatch to the right of each legend-name field) to pick a color from the palette manually (hex input also supported). If the

color-query command isn't configured, or the auto-read fails, GetSnap falls back to its default palette (CH1=yellow, CH2=green, etc., matching common oscilloscope conventions).

4.4 If a query fails

If reading the DIV conditions (voltage range / timebase / auto-color) fails, **the capture itself is never aborted**. An error message is shown, and the image is saved/copied without the DIV Conditions panel (the same appearance as when the feature is disabled).

What failed	Behavior
Voltage range / timebase	Error shown → suggests checking the commands in "Instrument Settings..."
Auto-color only	Error shown → suggests turning off auto-coloring and choosing colors manually
Both	Error shown → suggests checking both of the above
General communication error	Error detail shown

5. Supported Instruments

The actual commands can always be reviewed and edited from the dialog in "6. Adding & Editing Instruments" below.

Vendor	Image Format	Data Export	Notes
Tektronix	PNG	Yes	
Keysight	PNG	Yes	VNA (E5000 series) uses a separate preset
Yokogawa	PNG	Yes	Return-to-local uses GTL only (verified on real hardware, see below)
Rohde & Schwarz	PNG	Yes	Spectrum analyzer models (FSC/FSVA/FSWX) use separate presets
Rigol	BMP	Yes	Spectrum analyzer / VNA models use separate presets

6. Adding & Editing Instruments

Go to "Settings" → "Instrument Settings..." to add or edit supported instruments. Your edits persist across app restarts.

Save

Save Folder

C:\Users\daisu\GetSnap Captures

Browse...

Background Conversion Mode

Invert (B/W)

Clipboard

Clipboard Copy Image Size

100%

Hotkeys

Same actions are also available from the buttons in the "Capture" tab

☒ Enable hotkeys

Save to Folder

Ctrl+Alt+F1

Copy to Clipboard

Ctrl+W

Instruments

See Manual_ENG.pdf for the list of supported vendors and commands.

Instrument Settings...

Legend

Legend items (edit label / remove)

MODEL	
Ope. Mode	
BOM	
Condition	
Note	

+ Add Item

Legend Font

メイリオ (Meiryo)

15

Display

Language / 言語

English

Settings tab (Instrument Settings button)

6.1 First, find the model match string

Before adding a new instrument, connect to it once from the "Devices" tab. You can connect even without a registered preset — the ***IDN?** response string is shown as-is on screen (e.g. **Connected: ANRITSU,MS2840A,SN001,FW2.0 [Unsupported model (no preset registered)]**). Use part of that string (e.g. the vendor name) as the "model match string".

6.2 Editing an existing preset

Select an instrument from the list to open its form on the right. Edit the fields and press "Save" to override that instrument's settings only. Built-in presets cannot be renamed. "Reset to Default" restores the original content at any time.

The screenshot shows the 'Preset edit screen' for a Yokogawa instrument. On the left, under 'Registered instruments', a list includes Tektronix, Keysight VNA (E5000 Series), Keysight, Yokogawa (highlighted with a blue border), Rohde & Schwarz FSC, Rohde & Schwarz FSVA, Rohde & Schwarz FSX, and Rohde & Schwarz. A '+ Add New' button is at the bottom of the list. The right panel contains the following fields: a warning message 'This is a built-in preset. Its name cannot be changed (use "+ Add New" to register it under a different name. Editing and saving overrides this instrument's settings only.)'; a 'Name' field with 'Yokogawa'; a 'Model match string' field with 'yokogawa'; an 'Image save commands' field with ':IMAGE:FORMat PNG' and ':IMAGE:SEND?'; a 'Fallback commands' field with ':IMAGE:FORMat PNG' and ':IMAGE:SEND?'; and two empty fields for 'Image Format' and 'Delay Between Commands'.

Preset edit screen (Yokogawa example)

6.3 Adding a new instrument

Press "+ Add New" at the bottom left to open a blank form. Fill in the required fields (name, model match string, image save commands) and press "Save" to add it to the list. "Delete" removes it at any time.

The screenshot shows the 'Preset add-new screen' for an Anritsu instrument. On the left, under 'Registered instruments', a list includes Tektronix, Keysight VNA (E5000 Series), Keysight, Yokogawa, Rohde & Schwarz FSC, Rohde & Schwarz FSVA, Rohde & Schwarz FSX, and Rohde & Schwarz. A '+ Add New' button is at the bottom of the list. The right panel contains the following fields: a 'Name' field with 'Anritsu'; a 'Model match string' field with 'anritsu'; an 'Image save commands' field with ':HCOPY:SDUMP:DATA?'; a 'Fallback commands' field with 'Optional'; and two empty fields for 'Image Format' and 'Delay Between Commands'.

Preset add-new screen (Anritsu example)

6.4 Field Reference (basic)

Field	Description	Example
Name	Display name in the list (built-in presets cannot be renamed)	Anritsu
Model match string	Substring of the *IDN? response, matched case-insensitively; the first match from the top of the list is used	anritsu
Image save commands	Screen capture commands. One per line; the last line is the query that returns the image	:HCOPY:SDUMP:DATA?
Fallback commands	Used if the save commands fail. Leave blank to reuse the save commands	:DISPlay:DATA?
Image format	Image format of the returned data	PNG / BMP
Delay between commands	Wait time between multiple commands, in seconds	0.0–1.0
Data export commands (optional)	Waveform data retrieval commands	:WAVEform:SEND?
Data export format (optional)	Roughly, the saved file's extension	CSV
Return-to-local commands (optional)	Extra command(s) to restore the front panel after capture	Usually blank (see 6.5)
Notes (optional)	Free text	—

6.5 Leave "Return-to-local commands" blank in most cases

For GPIB/USB connections, GetSnap already restores the front panel automatically after capture via VISA-level GTL (Go To Local). Entering a command the instrument doesn't recognize here can actually cause a command-error message or leave the panel locked (**confirmed through real hardware testing on a Yokogawa DLM5058**). Only enter a command here — one you've verified against the model's programming manual — if GTL alone does not restore the panel for your instrument.

6.6 DIV Conditions commands (optional)

To use the "4.2 DIV Conditions" feature, the instrument's preset needs voltage range / timebase / (optionally) display-color query commands configured. A blank field simply means that part of the feature won't work — it has no effect on image capture itself.

Registered instruments

Tektronix
Keysight VNA (E5000 Series)
Keysight
Yokogawa
Rohde & Schwarz FSC
Rohde & Schwarz FSVA
Rohde & Schwarz FSWX
Rohde & Schwarz
+ Add New

Math trace voltage range query ({ch} is replaced with the trace number)
: MATH{ch}:VDIV?
Timebase query
: TIMEbase:TDIV?
Channel (CH) display color query (optional, {ch} is replaced with the channel number)
Optional. Falls back to the default color palette if blank or unreadable
Math trace display color query (optional, {ch} is replaced with the trace number)
Optional. Falls back to the default color palette if blank or unreadable
Notes (optional)
要検証：DLM/DLシリーズを想定した一般的な構成例。ハードコピー専用の白背景出力コマンドを確認できて
save_function_commandsとfallback_commandsは同一としている。機種別の programming マニュアルで
コマンドを確認する。
Save
Reset to Defau

DIV Conditions command fields

Field	Description	Example
Channel voltage range query	{ch} is replaced with the channel number	:CHANnel{ch}:VDIV?
Math trace voltage range query	{ch} is replaced with the trace number	:MATH{ch}:VDIV?
Timebase query	Retrieves the shared timebase	:TIMEbase:TDIV?
Channel display color query (optional)	{ch} is replaced with the channel number; the response can be hex, RGB, or a color name	:CHANnel{ch}:COLor?
Math trace display color query (optional)	Same as above, for Math traces	:MATH{ch}:COLor?

Display-color query commands are highly model-dependent, so they're left blank on the built-in presets. To use auto-coloring (see "4.3"), check your instrument's programming manual and enter the actual command here.

7. Troubleshooting

Symptom	Fix
Instrument doesn't appear in the list over GPIB	Install NI-VISA or Keysight IO Libraries Suite
Connects fine but capture fails	Check the failing step at the start of the error, then review commands in "6."
Want to change the image background color	Toggle "Background Conversion Mode" on the "Settings" tab

Symptom	Fix
Hotkey doesn't respond	May conflict with another app; use the buttons on the "Capture" tab instead
Front panel unresponsive after capture	See "6.5 Leave 'Return-to-local commands' blank in most cases"
DIV Conditions shows an error / doesn't appear	Check the command setup in "6.6 DIV Conditions commands". On error, the image is still saved without DIV Conditions
Channel colors don't match the real instrument	See "4.3 Choosing channel colors" — check the auto-color command setup, or choose colors manually